russbassett

Remote Quantitative Developer Assessment Test

Instructions: Precede each answer with the question you are answering.
Use acronyms only after you've explained them.
Use correct spelling and grammar.
Be sure to write your name.

Please submit your completed assessment as a PDF document.

***Candidate's Name*:**

PART ONE

1. Are you currently employed?

2. How many hours can you devote weekly to the Company?

3. Why do you think you fit the role well? Please provide examples of your skills and experiences that align with the requirements of this role.

4. Have you worked from home before?

5. What interests you about this role?

6. What motivates you?

7 Are you seeking employment in a company of a specific size, such as a small startup, medium-sized Company, or large corporation? Please explain your preference.

8 Describe your experience working with a culturally diverse group of people

9. a. Write a Python function that takes an array of real numbers and returns the maximum drawdown. Include a test case. Explain the time and space complexity of your solution above.

b. Given a large dataset of price ticks (timestamp, price), how would you efficiently compute the moving average for every 5-minute window? Provide a brief algorithm or code snippet.

10. a. Suppose you have two uncorrelated assets, A and B, with expected returns of 5% and 7%, and standard deviations of 10% and 15%, respectively.
What is the expected return and variance of a portfolio with 60% in A and 40% in B?

b. Describe how you would test whether a financial time series is stationary. What steps would you take if it is not?

PART TWO

1. Write the Black-Scholes formula for pricing a European call option. What are the key assumptions behind this model?

2: Share a challenging technical issue you've encountered and how you resolved it, focusing on your analytical and critical thinking skills.

3. How do you handle feedback and criticism from supervisors or colleagues?

4. What is your greatest strength? How will it help you in this role?

5. We want to understand your long-term career goals and how this position aligns with them. Please share your aspirations and how you see this role contributing to your career path.

6. What do you know about our Company?

7. How do you see yourself contributing to our Company's continued success and growth in this role?

8. In your opinion, what are the biggest challenges facing our Company in the market?

9. Discuss your familiarity with recent technological trends and how they can be leveraged within the industry.

10. You are given a new dataset containing minute-level trading data for a new asset class.

Describe your approach for cleaning, exploring, and building a predictive model based on this data.

PART THREE

1. What tools or strategies are most effective for communication and collaboration in a remote work environment?

2. If you were hiring for this role, which key personality traits and qualities would you look for?

3. How do you approach problem-solving as a team member versus an individual contributor?

4. How do you prioritize and manage your workload in a fast-paced environment?

5. What specific skills or experiences do you hope to gain from this role to help you achieve your career goals?

6. Imagine that you told a client you would be there at 10 am. It is now 10:30 am, and you won't finish your job until 11:30. You have a lunch meeting with another client at noon, followed by another job at 1:15 pm. How would you handle this situation?

7. In your experience, how frequently does the following problem occur: employees being afraid to express disagreement with their managers?

8. Your experience working on projects involving a consortium of companies is valuable. Please briefly share your experience in this area.

9. How much would you request per hour?

10. When would you be available to start if hired?

Technical Exercise

Overview:

As a Quantitative Developer at Russ Bassett Corporation, you will be expected to demonstrate strong quantitative analysis, data engineering, and coding skills. This exercise aims to assess your ability to work with real-world datasets, develop analytical models, and communicate actionable insights. You are encouraged to use any publicly available dataset of your choice, relevant to finance, economics, operations, or a field you find compelling.

Exercise Instructions:

Select a Real-World Dataset

- *Choose a publicly available dataset (e.g., from Kaggle, Quandl, FRED, Yahoo Finance, government open data portals, etc.).*
- *The dataset should be sufficiently large and complex to showcase your skills (a minimum of ~5,000 rows is recommended).*
- *Briefly describe the dataset, its source, and why you chose it.*

Define a Quantitative Problem

- Formulate a clear, data-driven question or problem relevant to the dataset.
- Examples: Forecasting, anomaly detection, portfolio optimization, pricing analysis, risk modeling, etc.
- State your objective and the potential value of solving this problem.

Data Exploration and Preprocessing

- Perform exploratory data analysis (EDA): summary statistics, visualizations, and initial insights.
- Explain and implement any data cleaning, feature engineering, or transformation steps.

Model Development and Analysis

- Select and implement at least one quantitative method or model (e.g., time series forecasting, regression, classification, clustering, simulation, etc.).
- Justify your choice of model(s).
- Evaluate model performance using appropriate metrics.

Results and Interpretation

- Present your findings with relevant visualizations and clear explanations.
- Discuss the practical implications of your analysis.
- Outline any assumptions, limitations, and potential next steps.

Code Submission

Submit a well-documented Jupyter notebook (or Python/R script) containing:

- ✓ *All code, outputs, and visualizations.*
- ✓ *Sufficient comments explaining your logic and process.*

Summary Report

Include a 1–2 page summary (markdown or PDF) outlining:

- ✓ *Dataset description and source.*
- ✓ *The problem statement.*
- ✓ *Key steps and results.*
- ✓ *Main insights and recommendations.*

Evaluation Criteria:

- ✓ *Technical correctness and rigor.*
- ✓ *Clarity of explanation and communication.*
- ✓ *Code readability and documentation.*
- ✓ *Creativity in problem formulation and analysis.*
- ✓ *Ability to draw actionable, data-driven conclusions.*

Submission:

- ✓ *Please submit your code (Jupyter notebook or script), summary report, and a link/source for your dataset.*

Optional: If you wish, host your completed notebook (e.g., on GitHub or Colab) and share the link in your submission.

We look forward to seeing your approach to quantitative problem-solving!

NOTE: You may attach your **portfolio or project samples** for the hiring panel to review.